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3 October 1986
SUPERSEDING
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28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTTOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, AWG size 20, class K stranding, uncoated (see note 1).
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation. Nominal wall thickness 0.020 inch. Telephone identification code applied by method 3.
Fourth - Two conductors shall be cabled together to form pairs, with a lay suitable to meet test requirements specified herein.
Fifth - The specified number of pairs (see table I) shall be cabled together in one or more layers, all in a left-hand direction. The length of lay shall be determined to permit the completed cable to meet the twisting and bending endurance tests. Fillers shall be used as necessary, to form a round cable. Cabling sequence shall be consecutive, starting with pair no. 1, from center outward. A single pair shall not be used as a central core for the cable.
Sixth - Binder tape applied helically with overlap.
Seventh - Cross-linked polyolefin jacket (see table I for thickness). Cable surface marking required.

NOTE: 1. At the manufacturer's option, conductors may consist of eight strands, each having a nominal diameter of 0.010 inch, around one strand having a nominal diameter of 0.0142 inch.

(B) denotes changes.

AMSC N/A FSC 6145
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TABLE I. Details.

Military part no. M24643/12	Type and size	Number of pairs	Cable jacket thickness nominal (inch)	Overall diameter maximum (inch)
-01UN	LSTTOP-3	3	0.060	0.480
-02UN	LSTTOP-5	5	.060	.590
-03UN	LSTTOP-10	10	.060	.700
-04UN	LSTTOP-15	15	.060	.830

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C, maximum..	11.53
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Insulation resistance - megohms/1000 feet, minimum.....	100
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Group B:	
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum.....	700
Elongation - percent, minimum.....	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Twisting and bending endurance - (see note 2)	
At 75 ± 2°C, cycles, minimum.....	6000
At minus 30 ± 2°C, cycles, minimum.....	2000

NOTE: 2. Roller diameter for bending endurance test shall be as follows:

Cable type and size	Roller diameter (inches, maximum)
LSTTOP-3	1.4
LSTTOP-5	1.8
LSTTOP-10	2.1
LSTTOP-15	3.0

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Aging and compatibility (cable)(125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
(B) Insulation	45

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3, 5 and 10	2500
15	1500

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
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